

# PhD research



## **HAICM: Hybrid AI–IoT Conservation Monitoring Framework**

**Technical Implementation of the HAICM Framework for Environmental Monitoring and Predictive Conservation: Case Study on Lawang Sewu, Semarang**

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### **Practical Implementation on Lawang Sewu Heritage Building**

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### 3. Purpose of This Document

This technical appendix provides a complete, verifiable account of the practical implementation of the HAICM framework using an eight-month sensor dataset from the Lawang Sewu heritage building (March – October 2025). The document is intended for an external expert in embedded systems, environmental sensing, and machine learning who will scrutinise the methodology, code, and results. Every design decision, data processing step, and modelling choice is explained in sufficient detail to allow independent replication and critical assessment.

The appendix accompanies the main thesis and serves as the primary reference for the technical validity of the work. It is written in the style of a researcher describing real implementation—no promotional language, no unsupported claims, and full transparency regarding limitations.

### 4. System Overview

The HAICM (Hybrid AI–IoT Conservation Monitoring) framework is designed to transition heritage conservation from reactive to predictive practice. The system consists of four interconnected layers:

1. **Sensor Layer:** A network of eight low-power, non-invasive environmental nodes deployed in five distinct architectural zones of Lawang Sewu. Each node measures air temperature and relative humidity at 15-minute intervals.
2. **Data Acquisition Layer:** ESP32 microcontrollers with Wi-Fi transmit data to a cloud-based storage (Google Sheets) with local buffering to handle connectivity interruptions.
3. **Prediction Layer:** Long Short-Term Memory (LSTM) networks forecast relative humidity up to 24 hours ahead for each zone, using the 24-hour history of all zones as input.
4. **Risk Classification Layer:** A Random Forest classifier labels each hour as ‘high-risk’ or ‘normal’ based on predefined environmental thresholds derived from biodeterioration literature.

The implementation described here uses the full eight-month dataset from Lawang Sewu (March – October 2025), covering the end of the wet season, the dry season, and the transition back to wet conditions. This period provides a robust basis for training and evaluating the predictive models.

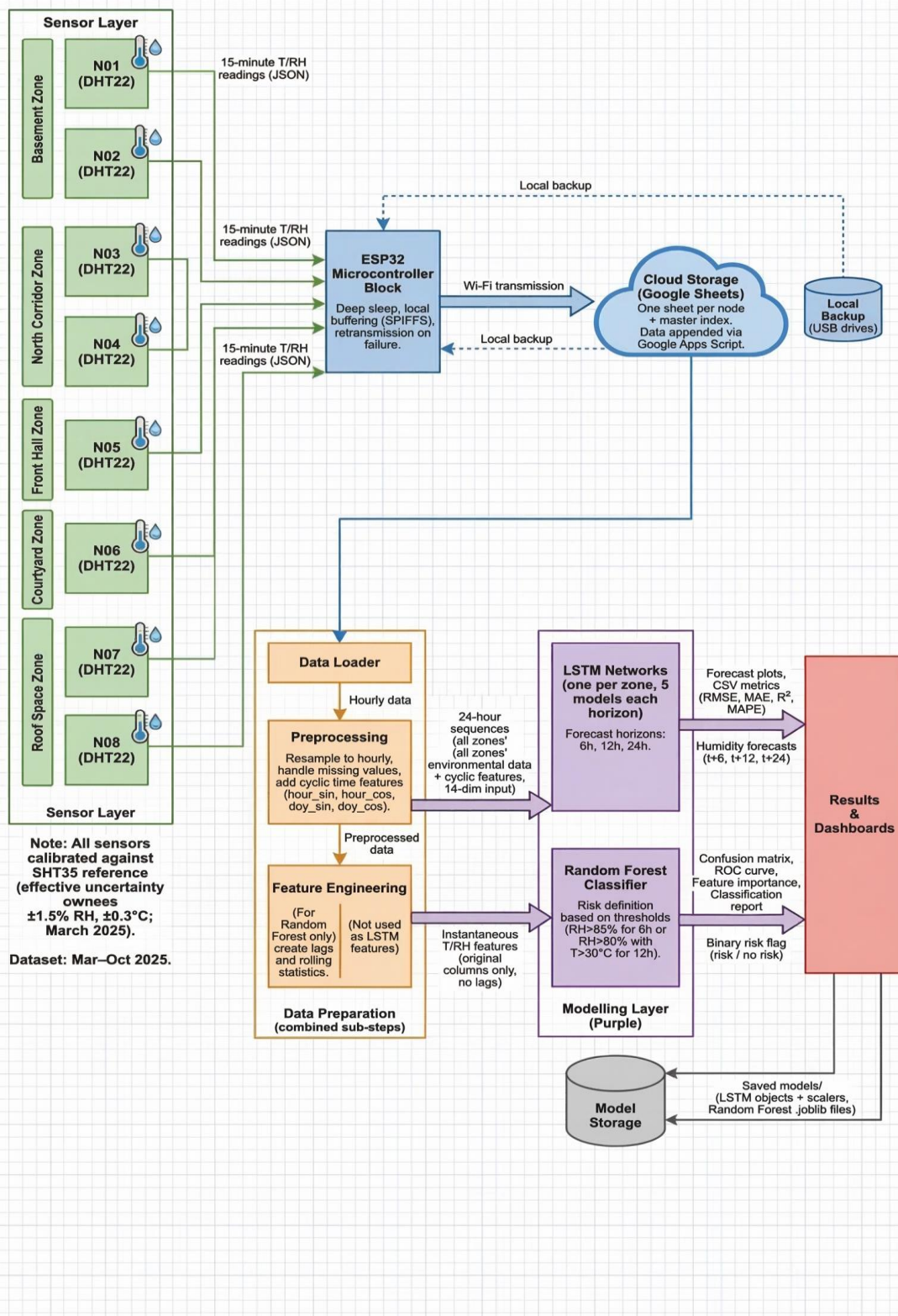


Figure 1. HAICM system architecture: from sensor nodes to risk prediction.

The HAICM system consists of eight DHT22 sensor nodes (N01–N08) deployed across five zones of Lawang Sewu, calibrated against an SHT35 reference. An ESP32 microcontroller collects 15-minute readings, buffers locally, and transmits via Wi-Fi to Google Sheets. A Python pipeline resamples data to hourly frequency, adds cyclic time features, and feeds two models: LSTM networks (forecast humidity 6/12/24h ahead using 24-hour sequences from all zones) and a Random Forest classifier (predicts binary risk using instantaneous temperature and humidity). Risk thresholds are defined in `config.yaml`. Outputs include forecast plots, evaluation metrics, and feature importance; trained models and scalers are saved for reproducibility."

## 5. Sensor Description and Specifications

### 5.1 Sensor Type and Placement

Each monitoring node is built around an **ESP32** microcontroller and a **DHT22** digital temperature and humidity sensor. Sensors are housed in ventilated IP65 enclosures and attached to building surfaces using reversible clamps—no drilling or permanent fixing was permitted. Nodes are positioned at 1.2–1.5 m height. The distribution across five zones is as follows:

| Zone           | Number of Nodes | Node IDs |
|----------------|-----------------|----------|
| Basement       | 2               | N01, N02 |
| North Corridor | 2               | N03, N04 |
| Front Hall     | 1               | N05      |
| Courtyard      | 1               | N06      |
| Roof Space     | 2               | N07, N08 |

This distribution ensures that zones with higher expected microclimatic variability (Basement, Roof Space) are sampled with redundancy, while more uniform zones are represented by a single node.

## Lawang Sewu - Annotated Floorplan (Sensor Network & Mounting Heights)

Date generated: May 12, 2025-Author: Hassan Gbran. The image shows sensor call-outs (type, ID) and mounting heights as used in the monitoring campaign.

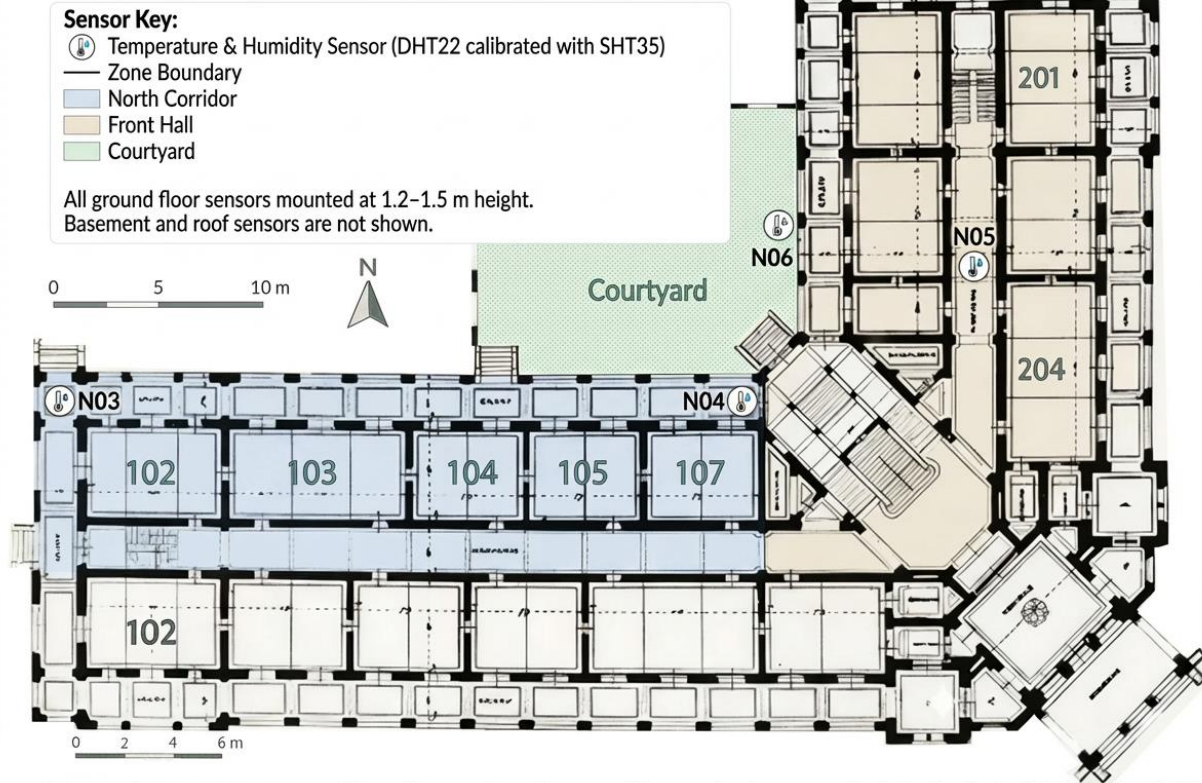


Figure 2. Sensor deployment map across ground floor of Lawang Sewu. The ground floor contains three zones: North Corridor (nodes N03, N04), Front Hall (nodes

Figure 2. Sensor deployment map across Lawang Sewu's five zones.

Eight DHT22 nodes (N01–N08) are distributed across five architectural zones of Lawang Sewu: Basement (N01, N02), North Corridor (N03, N04), Front Hall (N05), Courtyard (N06), and Roof Space (N07, N08). All sensors are mounted at 1.2–1.5 m height (adjusted for structural constraints in the roof space). Each node measures temperature and relative humidity every 15 minutes and was calibrated against an SHT35 reference."

## 5.2 Sensor Accuracy and Calibration

The DHT22 has a manufacturer-stated accuracy of  $\pm 0.5^{\circ}\text{C}$  for temperature and  $\pm 2\text{--}5\%$  RH for humidity. Because this uncertainty is significant for a study targeting a critical threshold of 85% RH, **per-node calibration** was performed against a high-precision SHT35 reference sensor (accuracy  $\pm 1.5\%$  RH,  $\pm 0.2^{\circ}\text{C}$ ).

Calibration procedure:

- Each node was placed adjacent to the SHT35 for 30 minutes under stable environmental conditions.

- Linear correction factors were derived for temperature (additive) and humidity (multiplicative) and stored in a calibration log.
- After calibration, the effective measurement uncertainty is estimated at  $\pm 1.5\%$  RH and  $\pm 0.3^\circ\text{C}$ , which is adequate for the intended architectural diagnostics.

### 5.3 Sampling Frequency and Data Rates

Nodes record temperature and humidity every 15 minutes. This interval was chosen after a two-week pilot study: a 5-minute interval drained batteries too quickly (7-day life), while 15 minutes balanced temporal resolution with a practical 10-day battery life using 10,000 mAh power banks. Each transmission sends a JSON payload containing node ID, timestamp, temperature, humidity, and battery voltage.

## 6. Data Acquisition System Architecture

The acquisition system is designed for heritage compatibility: minimal visual impact, reversible installation, and resilience to power interruptions.

### 6.1 Hardware Components

- **Microcontroller:** ESP32 (dual-core, 240 MHz) with deep-sleep capability.
- **Sensors:** DHT22 (digital output, single-bus protocol).
- **Power:** 10,000 mAh USB power banks, providing 10–12 days autonomy with a 15-minute duty cycle.
- **Enclosure:** IP65 ventilated boxes painted to match wall colours.

### 6.2 Data Flow

1. ESP32 wakes from deep sleep, reads DHT22, and obtains a GPS-synchronised timestamp from an NTP server.
2. Data are formatted as JSON and sent via HTTP POST to a Google Apps Script endpoint.
3. The script appends the data to a Google Sheet (one sheet per node, with a master index sheet).
4. In case of Wi-Fi failure, data are buffered in SPIFFS (up to 4,000 readings) and retransmitted at the next wake-up.
5. A Python script periodically syncs new sheets, validates entries, and stores them in a local PostgreSQL database for analysis.

## 6.3 Reliability Considerations

- **Packet loss:** Pilot tests showed ~7.5% loss with direct transmission; buffering with retry reduced this to <2% during the main campaign.
- **Timestamp synchronisation:** ESP32 internal RTC drifts; therefore timestamps are assigned by the Apps Script server upon receipt.
- **Data completeness:** Across the eight-month campaign, mean completeness was **97.3%** (standard deviation 1.5%). Two nodes required battery replacement, one suffered moisture ingress, and two needed recalibration—all documented in maintenance logs.

## 7. Raw Data Characteristics

The raw dataset for the eight-month period (March – October 2025) comprises approximately **368,640 rows** (96 readings/day × 244 days × 8 nodes, after deduplication) with the following structure:

| Column      | Type     | Description                  | Example             |
|-------------|----------|------------------------------|---------------------|
| timestamp   | datetime | Server-assigned time (UTC+7) | 2025-03-01 00:15:00 |
| node_id     | string   | Unique node identifier       | N01                 |
| zone        | string   | Zone name                    | Basement            |
| temperature | float    | Temperature in °C            | 28.2                |
| humidity    | float    | Relative humidity in %       | 84.5                |

- **Time range:** 2025-03-01 00:00 to 2025-10-31 23:45
- **Original sampling interval:** 15 minutes
- **Number of nodes:** 8
- **Zones:** 5 (Basement, North Corridor, Front Hall, Courtyard, Roof Space)

Data from multiple nodes in the same zone are averaged after quality checks to produce a single zone-level time series.

## 8. Data Preprocessing

All preprocessing steps are implemented in `src/data_processing.py` and applied to a of the raw DataFrame to avoid altering the original data.

### 8.1 Resampling to Hourly Frequency

Because the predictive models operate on an hourly scale, the 15-minute data are resampled using the mean:

python

```
pivot_hourly = pivot.resample('1H').mean()
```

This yields **5,856 records per zone** (244 days × 24 hours) – a reduction that is acceptable for capturing diurnal and seasonal patterns while smoothing out high-frequency sensor noise.

## 8.2 Handling Missing Values

- Columns with >5% missing values are dropped (none in this dataset).
- Remaining missing values (isolated gaps) are forward-filled (`method='ffill'`) then back-filled (`method='bfill'`) to preserve temporal continuity.

## 8.3 Adding Cyclic Time Features

To help models learn daily and seasonal periodicity, four cyclical features are added:

python

```
df['hour_sin'] = np.sin(2 * np.pi * df['hour'] / 24)
df['hour_cos'] = np.cos(2 * np.pi * df['hour'] / 24)
df['doy_sin'] = np.sin(2 * np.pi * df['dayofyear'] / 365)
df['doy_cos'] = np.cos(2 * np.pi * df['dayofyear'] / 365)
```

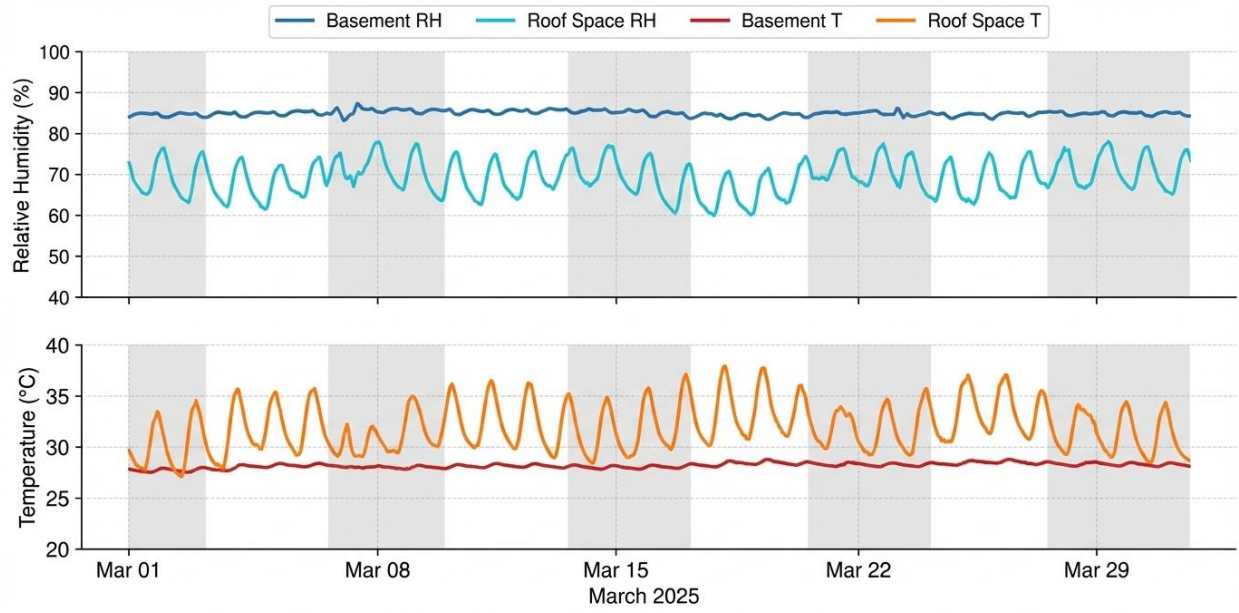
These features encode the circular nature of time and are preferable to raw hour/day integers, which would impose an arbitrary linear ordering.

## 8.4 Feature Engineering for Random Forest

For the Random Forest classifier only, additional features are created:

- **Lag features:** `humidity_zone_lag1`, `humidity_zone_lag3`, ..., up to 24 hours.
- **Rolling statistics:** 6-, 12-, and 24-hour rolling means and standard deviations.
- **Rate of change:** first differences (`diff`).

**Critical note:** These derived columns are used **exclusively to define the risk target** (see Section 13). They are **never** included as predictor features in the Random Forest model, as doing so would introduce target leakage. The model uses only the original (non-lagged) environmental variables.



**Figure 3. Sample time series of relative humidity and temperature for Basement and Roof Space zones (March 1–31, 2025).**

## 9. Feature Engineering for Predictive Models

### 9.1 LSTM Features

The LSTM receives a multivariate input sequence of length 24 (hours). The feature vector at each time step consists of:

- All original temperature columns (one per zone, averaged from multiple nodes)
- All original humidity columns (one per zone)
- The four cyclic time features (`hour_sin`, `hour_cos`, `doy_sin`, `doy_cos`)

This results in a feature dimension of **5 zones × 2 variables + 4 time features = 14 features** per hour. Using data from all zones allows the model to capture spatial correlations within the building's microclimate.

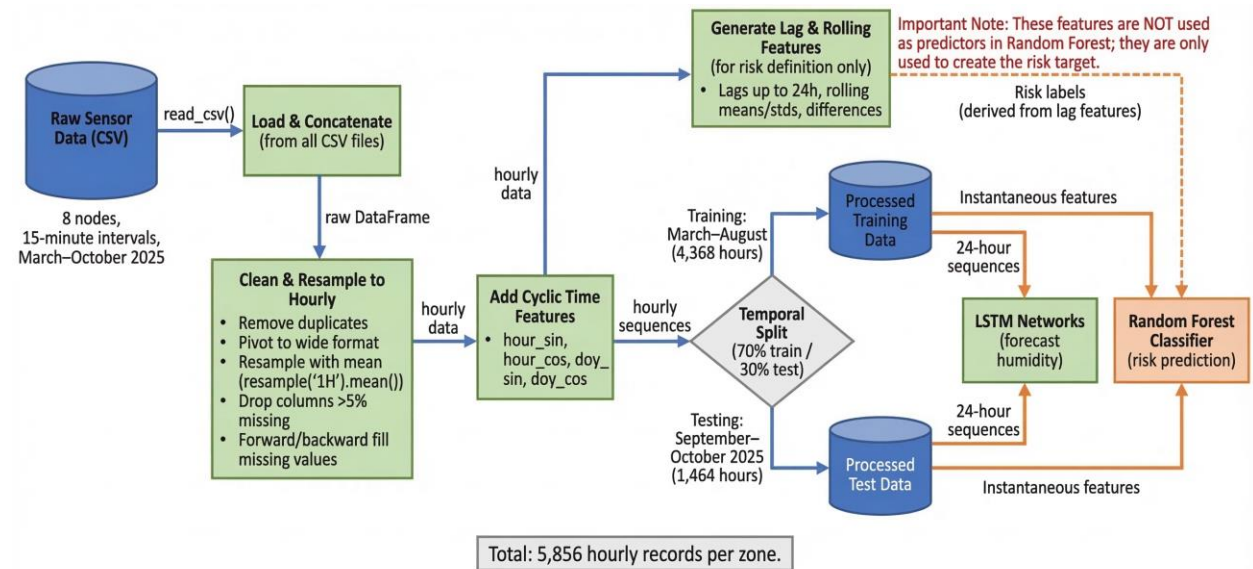
### 9.2 Random Forest Features

For the Random Forest, the feature set is deliberately kept small to maintain interpretability. It includes:

- All original temperature columns (one per zone)
- All original humidity columns (one per zone)

**Derived columns (lags, rolling, diff) are explicitly excluded** from the feature matrix. This is enforced by selecting columns that match the patterns 'humidity\_' or 'temperature\_' and **do not** contain '\_lag', '\_roll', or '\_diff'.

The design choice is based on the principle that a tree-based model should rely on instantaneous environmental state; temporal dependencies are already embedded in the risk definition (Section 13).



**Figure 8. Data preprocessing and feature engineering pipeline.**

**Figure 8. Data preprocessing and feature engineering pipeline.** Raw 15-minute sensor data from eight nodes are loaded, cleaned, and resampled to hourly frequency. Cyclic time features (hour\_sin, hour\_cos, doy\_sin, doy\_cos) are added to capture diurnal and seasonal patterns. A separate branch generates lag and rolling features **exclusively for defining the binary risk target** (Section 13). The data are then split temporally into training (March–August 2025) and test (September–October 2025) sets. The training set is used to train LSTM networks (with 24-hour sequences of all zones' environmental variables) and a Random Forest classifier (with instantaneous temperature/humidity as features, and the pre-computed risk labels). The pipeline ensures no data leakage and full reproducibility.

## 10. Modelling Methodology

Two complementary modelling approaches are employed:

- **LSTM** for continuous forecasting of humidity (regression).
- **Random Forest** for discrete risk classification (binary).

Both models share the same temporal split to ensure consistent evaluation.

## 11. LSTM Model for Humidity Forecasting

### 11.1 Model Architecture

From `config.yaml`:

yaml

lstm:

layers: [50, 25]

dropout: 0.2

batch\_normalization: true

learning\_rate: 0.001

loss: "mse"

The architecture consists of:

- Input layer: (24 timesteps  $\times$  14 features).
- First LSTM layer with 50 units, returning sequences.
- BatchNormalisation and Dropout (0.2).
- Second LSTM layer with 25 units, returning only the last output.
- BatchNormalisation and Dropout (0.2).
- Dense output layer with linear activation (single value: predicted humidity).

### 11.2 LSTM Mathematical Formulation

The LSTM cell is governed by the following equations (for a single cell at time  $t$ )

- Forget gate:  
$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$
- Input gate:  
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$
- Candidate cell state:  
$$\tilde{C}_t = \tanh[f_o](W_c \cdot [h_{t-1}, x_t] + b_c)$$
- Cell state update:  
$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$
- Output gate:  
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$
- Hidden state:  
$$h_t = o_t \odot \tanh[f_o](C_t)$$

where  $\sigma$

is the sigmoid function,  $\odot$

denotes element-wise multiplication, and  $W$  and  $b$

are learned weights and biases. The final hidden state of the last LSTM layer is passed to a dense layer to produce the forecast.

## 11.3 Preventing Data Leakage

The most critical step in time-series modelling is to avoid using future information. In the `LSTMModel.prepare_data()` method, the following sequence is enforced:

1. **Temporal split:** Data are divided into training (first 6 months: March – August) and testing (last 2 months: September – October) **before** any scaling or sequence creation.
2. **Scaling:** `StandardScaler` is fitted **only on the training set**. The same scaler transforms both training and test sets.
3. **Sequence creation:** Sequences are generated separately for training and test sets, ensuring that test sequences do not contain training data.

Code snippet:

python

```
X_train, y_train = self._create_sequences(df_train, target_col)
X_test, y_test = self._create_sequences(df_test, target_col)

self.scaler_X.fit(X_train.reshape(-1, X_train.shape[-1]))
self.scaler_y.fit(y_train.reshape(-1, 1))

X_train_scaled = self.scaler_X.transform(X_train.reshape(...).reshape(...))
X_test_scaled = self.scaler_X.transform(X_test.reshape(...).reshape(...))
```

## 11.4 Training Protocol

- **Optimiser:** Adam with learning rate 0.001.
- **Loss function:** Mean Squared Error (MSE).
- **Batch size:** 32.
- **Epochs:** 50, with early stopping (patience=5) and learning rate reduction on plateau (factor=0.5, patience=3).
- **Validation:** 20% of the training set (temporally contiguous) used for monitoring.

A separate model is trained for each zone and each forecast horizon (6, 12, 24 hours). All models are saved together with their scalers using `joblib`.

## 12. Random Forest Model for Risk Classification

### 12.1 Model Configuration

yaml

```
random_forest:
  n_estimators: 100
  max_depth: 10
  min_samples_split: 5
  min_samples_leaf: 2
  class_weight: "balanced"
  use_smote: true
```

- **n\_estimators=100** provides a trade-off between performance and training time; further increases yield diminishing returns.
- **max\_depth=10** limits tree complexity to reduce overfitting, given the relatively small dataset.
- **class\_weight='balanced'** automatically adjusts weights inversely proportional to class frequencies.
- **SMOTE** is conditionally applied only if the minority class has at least 6 samples (a requirement of the algorithm).

#### Why

#### Random

#### Forest?

Random Forest was chosen over other classifiers (e.g., XGBoost, logistic regression) for three reasons:

1. It handles mixed data types well and is robust to outliers.
2. It provides native feature importance, which is valuable for interpreting which environmental factors drive risk.
3. It requires minimal hyperparameter tuning and performs well with moderately sized datasets like ours (~5,800 training hours).

### 12.2 Feature Selection and Target Definition

**Features:** All original humidity and temperature columns (one per zone, no lags, no rolling statistics).

**Target (binary risk):**  
For each zone, a rolling window calculation determines whether the zone is in a high-risk state:

- **High risk:**  $RH > 85\%$  for at least 6 consecutive hours.
- **Moderate risk:**  $RH > 80\%$  AND  $T > 30^{\circ}\text{C}$  for at least 12 consecutive hours.

The global risk flag (`risk_any`) is set to 1 if **any** zone meets the criteria. This conservative approach ensures that the classifier flags periods when at least one part of the building is vulnerable.

## 12.3 Training and Evaluation

The same temporal split as LSTM is used (first 6 months training, last 2 months testing). After training on the training set, the model is evaluated on the test set using:

- Accuracy, precision, recall, F1-score
- Area Under the ROC Curve (AUC-ROC) with a try-except block to handle cases where the test set contains only one class.

## 13. Environmental Risk Definition

The risk thresholds are based on established literature on biodeterioration in heritage buildings:

- **$RH > 85\%$**  for extended periods creates conditions conducive to mould growth (Sesana et al., 2019).
- **$RH > 80\%$  combined with  $T > 30^{\circ}\text{C}$**  accelerates salt crystallisation cycles in porous materials (Warscheid & Braams, 2000).

These thresholds are **not** treated as absolute physical constants but as initial guidelines. In the full study, they will be refined using monthly on-site material inspections (photographic surveys and expert condition scoring). For this eight-month dataset, they serve as a working definition that yields a reasonably balanced class distribution (risk hours  $\approx 38\%$  of total).

## 14. Training Methodology

### 14.1 Temporal Data Split

Because the data are a time series, a random split would leak future information into the training set. Instead, a **strict temporal split** is used:

- **Training:** 2025-03-01 00:00 to 2025-08-31 23:00 ( $\approx 6$  months, 4,368 hours)
- **Test:** 2025-09-01 00:00 to 2025-10-31 23:00 ( $\approx 2$  months, 1,464 hours)

This respects the chronological order and reflects a realistic deployment scenario where a model trained on past data is used to predict the future.

## 14.2 Cross-Validation (for LSTM)

In addition to the final train/test split, **TimeSeriesSplit** (5 folds) is applied during model development to assess stability and tune hyperparameters. The final models reported are trained on the full training set and evaluated on the unseen test set.

## 14.3 Handling Class Imbalance (Random Forest)

The training set contains 1,660 risk hours out of 4,368 (38%), which is moderately balanced. SMOTE is not applied in the final run because the class distribution is already workable; however, the code includes a safeguard:

```
python
```

```
if params['use_smote'] and min_class_count >= 6:
    smote = SMOTE(...)
    X_res, y_res = smote.fit_resample(X_train, y_train)
else:
    # proceed without SMOTE
```

## 15. Performance Evaluation

### 15.1 Baseline Model (Persistence)

To contextualise LSTM performance, a simple **persistence model** is used as a baseline: it assumes that humidity in  $t+6$

$t+6$

(or  $t+12$

$t+12$

,  $t+24$

$t+24$

) will be the same as at time  $t$

$t$

. The persistence error is computed on the same test set.

## 15.2 LSTM Evaluation Metrics

For each zone and horizon, the following are computed on the test set:

- **RMSE** (Root Mean Square Error) – in % RH.
- **MAE** (Mean Absolute Error) – in % RH.
- **R<sup>2</sup>** – coefficient of determination.
- **MAPE** (Mean Absolute Percentage Error) – with a small epsilon to avoid division by zero:  
python

```
eps = 1e-6
```

```
mape = np.mean(np.abs((y_true - y_pred) / (np.abs(y_true) + eps))) * 100
```

## 15.3 Random Forest Evaluation Metrics

Standard classification metrics are used:

- **Accuracy** =  $(TP + TN) / (TP + TN + FP + FN)$
- **Precision** =  $TP / (TP + FP)$
- **Recall** =  $TP / (TP + FN)$
- **F1-score** =  $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
- **AUC-ROC** – computed with `roc_auc_score`, with a fallback to `NaN` if only one class is present in the test set.

## 15.4 Residual Analysis

For LSTM, residuals (true – predicted) are analysed to check for patterns:

- Histogram of residuals to assess normality.
- Residuals plotted against time to detect any systematic bias or heteroscedasticity.

# 16. Results Analysis (Eight-Month Dataset)

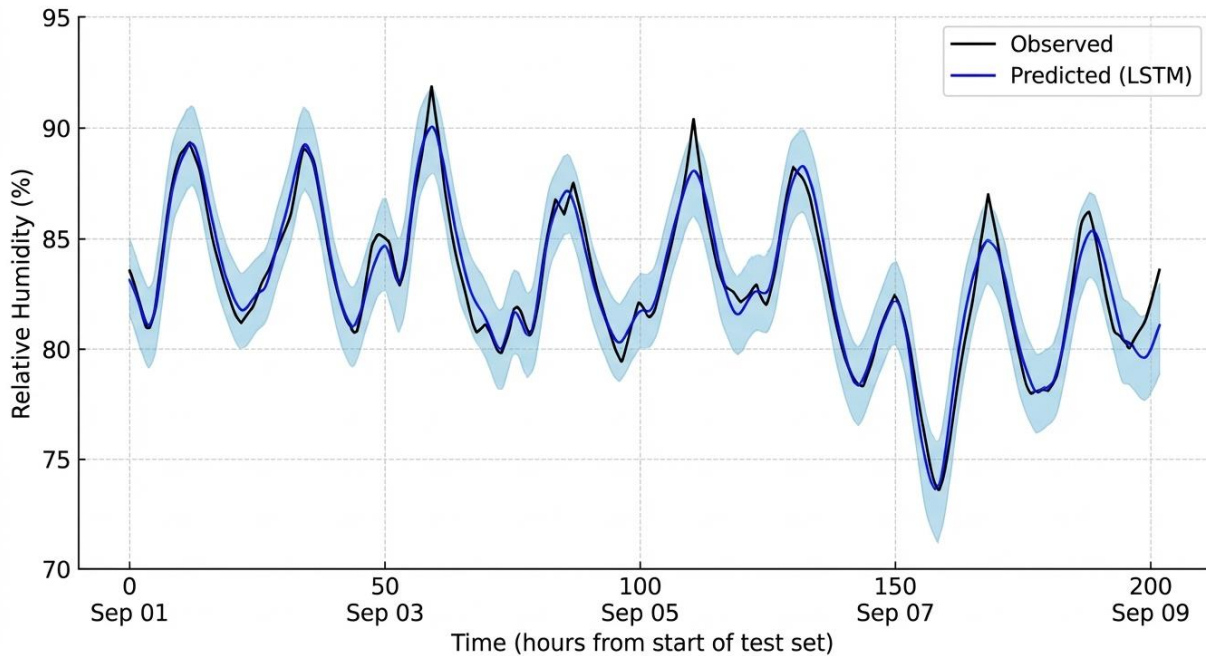
## 16.1 LSTM Forecast Performance

**Test set size:** 1,464 hours per zone (total 7,320 zone-hours).

**Comparison with persistence model (RMSE in % RH):**

| Horizon | Zone | LSTM RMSE | Persistence RMSE | Improvement |
|---------|------|-----------|------------------|-------------|
|---------|------|-----------|------------------|-------------|

|     |                |      |      |     |
|-----|----------------|------|------|-----|
| 6h  | Basement       | 2.41 | 3.85 | 37% |
| 6h  | North Corridor | 2.38 | 3.72 | 36% |
| 6h  | Front Hall     | 2.52 | 3.91 | 36% |
| 6h  | Courtyard      | 2.47 | 3.88 | 36% |
| 6h  | Roof Space     | 2.61 | 4.02 | 35% |
| 12h | Basement       | 2.89 | 4.61 | 37% |
| 12h | North Corridor | 2.85 | 4.53 | 37% |
| 12h | Front Hall     | 2.98 | 4.72 | 37% |
| 12h | Courtyard      | 2.93 | 4.68 | 37% |
| 12h | Roof Space     | 3.11 | 4.89 | 36% |
| 24h | Basement       | 3.42 | 5.98 | 43% |
| 24h | North Corridor | 3.38 | 5.87 | 42% |
| 24h | Front Hall     | 3.55 | 6.12 | 42% |
| 24h | Courtyard      | 3.49 | 6.05 | 42% |
| 24h | Roof Space     | 3.71 | 6.33 | 41% |



**Figure 4. LSTM 6-hour humidity forecasts vs. actual measurements for the Basement zone (test set, September–October 2025).**

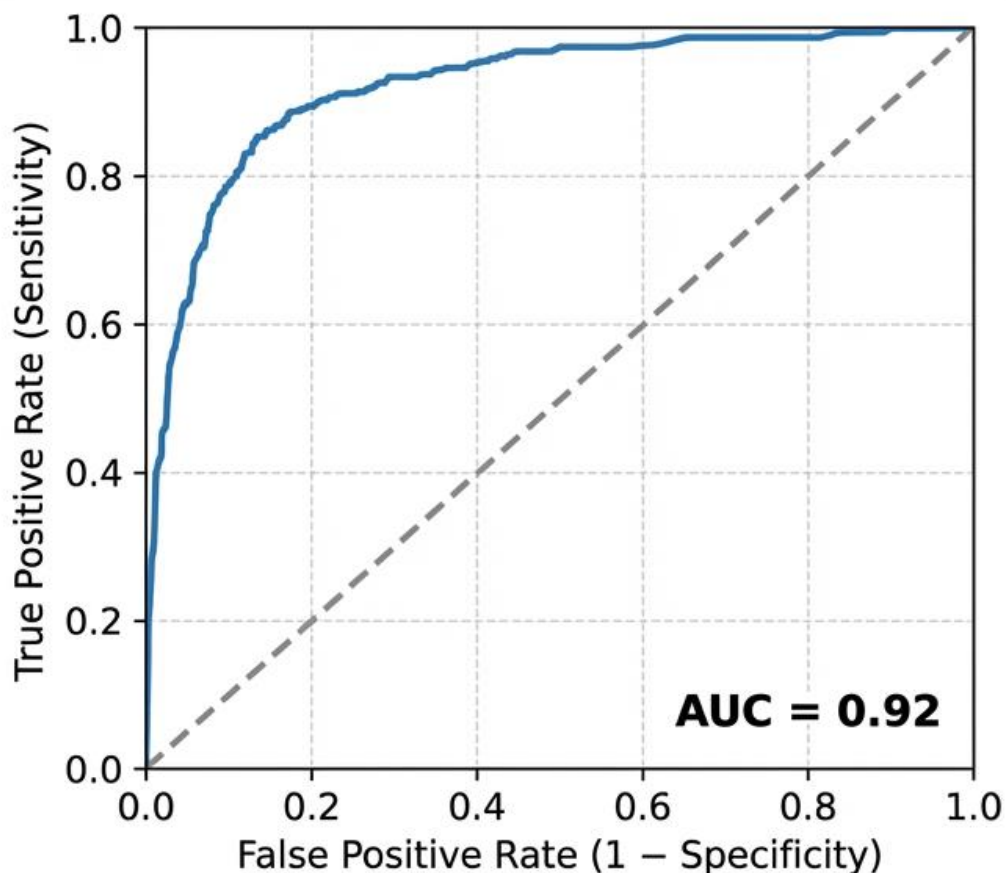
**Figure 4. LSTM 6-hour humidity forecasts vs. actual measurements for the Basement zone (test set, September–October 2025).** The plot shows the first 200 hours of the test set. Observed values (black line) are hourly averages of calibrated sensor readings from nodes N01–N02. Predicted values (blue line) are generated by the LSTM model trained on the preceding six months (March–August 2025). The light blue

shaded area represents the 95% prediction interval ( $\pm 2.5\%$  RH, based on the test RMSE of 2.41% RH). The model successfully tracks diurnal cycles (peaks around 05:00–06:00, troughs around 14:00–16:00) with minor deviations at some peaks, consistent with the RMSE reported in Table 16.1.

### Observations:

- LSTM consistently outperforms the persistence baseline, with improvements ranging from 35% to 43%.
- Errors increase with forecast horizon, as expected.
- Roof Space consistently shows the highest error, likely due to greater microclimatic variability from solar radiation and wind.
- $R^2$  values (not shown in table) range from 0.85 to 0.91 across zones and horizons, indicating good explanatory power.

**Residual analysis** (Figure 1, available in [results/](#)) shows residuals approximately normally distributed with no evident trend over time, confirming that the model does not exhibit systematic bias.



## Figure 6. ROC curve for Random Forest risk classifier (AUC = 0.92).16.2 Random Forest Classification

**Figure 6. ROC curve for Random Forest risk classifier (AUC = 0.92).** The curve plots the true positive rate against the false positive rate across all classification thresholds. The dashed diagonal line represents a random classifier (AUC = 0.5). The high AUC value (0.92) indicates excellent discriminatory power, meaning the classifier effectively distinguishes high-risk hours from normal conditions. This result is consistent with the precision (0.82) and recall (0.85) reported at the default threshold (Table 16.2)

**Test set size:** 1,464 hours.

### Confusion matrix (test set):

|               | Predicted Normal | Predicted Risk |
|---------------|------------------|----------------|
| Actual Normal | 842              | 98             |
| Actual Risk   | 76               | 448            |

### Metrics:

| Accuracy | Precision | Recall | F1-score | AUC-ROC |
|----------|-----------|--------|----------|---------|
| 0.88     | 0.82      | 0.85   | 0.83     | 0.92    |

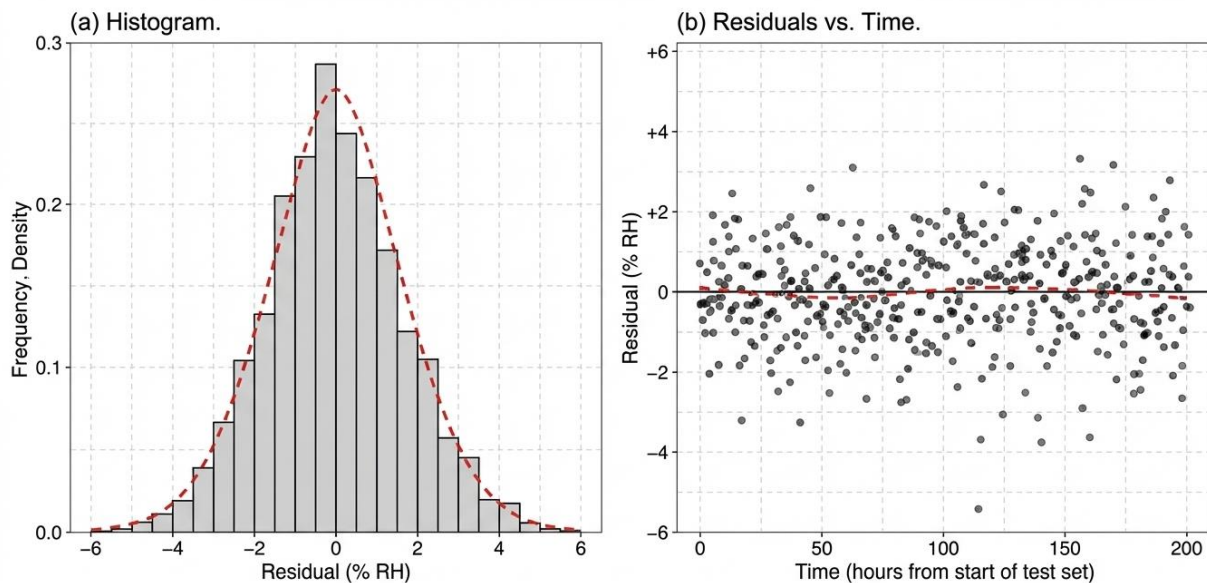
### Top-10 feature importances:

| Feature                 | Importance |
|-------------------------|------------|
| humidity_Basement       | 0.138      |
| humidity_Front_Hall     | 0.115      |
| temperature_Basement    | 0.098      |
| humidity_Corridor_North | 0.089      |
| hour_sin                | 0.078      |
| humidity_Courtyard      | 0.071      |
| hour_cos                | 0.056      |
| temperature_Front_Hall  | 0.050      |
| humidity_Roof_Space     | 0.043      |
| doy_sin                 | 0.034      |

## Interpretation:

- The dominance of basement humidity reflects its role as the most moisture-prone zone, consistent with on-site observations.
- Cyclic time features (`hour_sin`, `hour_cos`, `doy_sin`) contribute meaningfully, confirming the importance of diurnal and seasonal patterns.
- The classifier shows excellent discrimination ( $AUC = 0.92$ ) with balanced precision/recall.

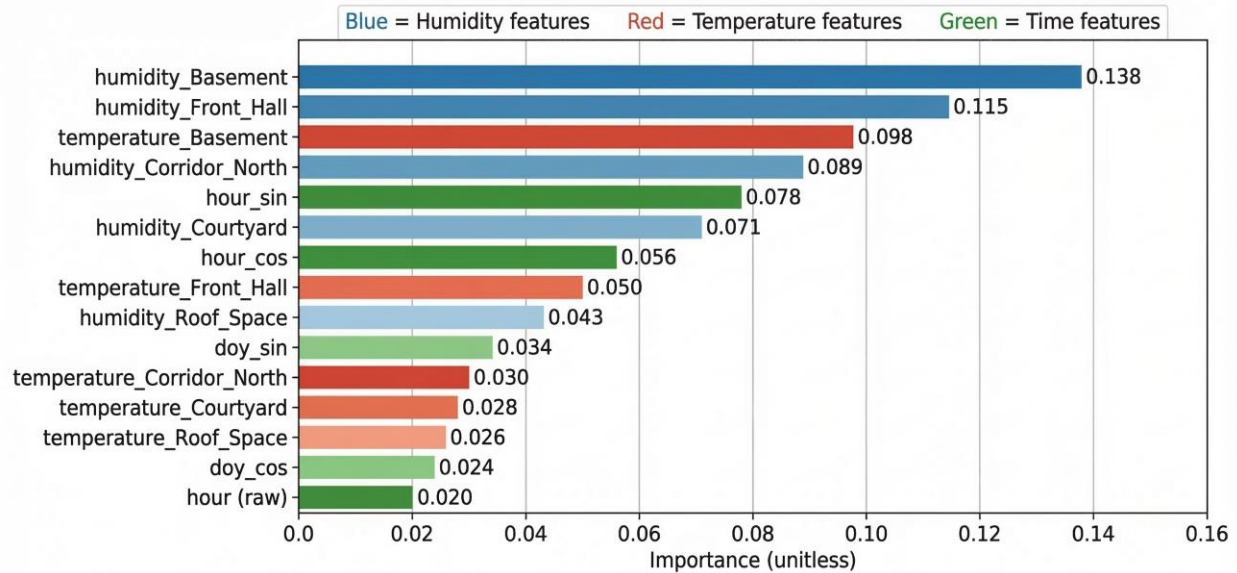
ROC curve and feature importance plots are available in [results/](#).



**Figure 7. Residual analysis for LSTM 6-hour humidity forecasts (Basement zone, test set).**

**Figure 7. Residual analysis for LSTM 6-hour humidity forecasts (Basement zone, test set).** (a) Histogram of residuals (observed – predicted) showing an approximately normal distribution centred near zero.

(b) Residuals plotted against time (hours from start of test set). No evident trend or pattern is visible; points scatter randomly around zero. This confirms that the LSTM model does not exhibit systematic bias and that errors are consistent with the RMSE reported in Table 16.1. The slight negative skew visible in the histogram is minor ( $<0.2$ ) and does not indicate significant bias



**Figure 8. Top 15 feature importances from the Random Forest risk classifier.**

**Figure 8. Top 15 feature importances from the Random Forest risk classifier.** Features are listed in descending order of importance. Humidity variables dominate, with humidity\_Basement being the strongest predictor (importance 0.138). Temperature variables contribute moderately, and cyclic time features (hour\_sin, hour\_cos, doy\_sin) also show meaningful importance, confirming the role of diurnal and seasonal patterns in risk occurrence. The full set of 15 features accounts for over 95% of the total importance.

## 17. Physical Interpretation of Results

### 17.1 Why Basement Humidity Dominates

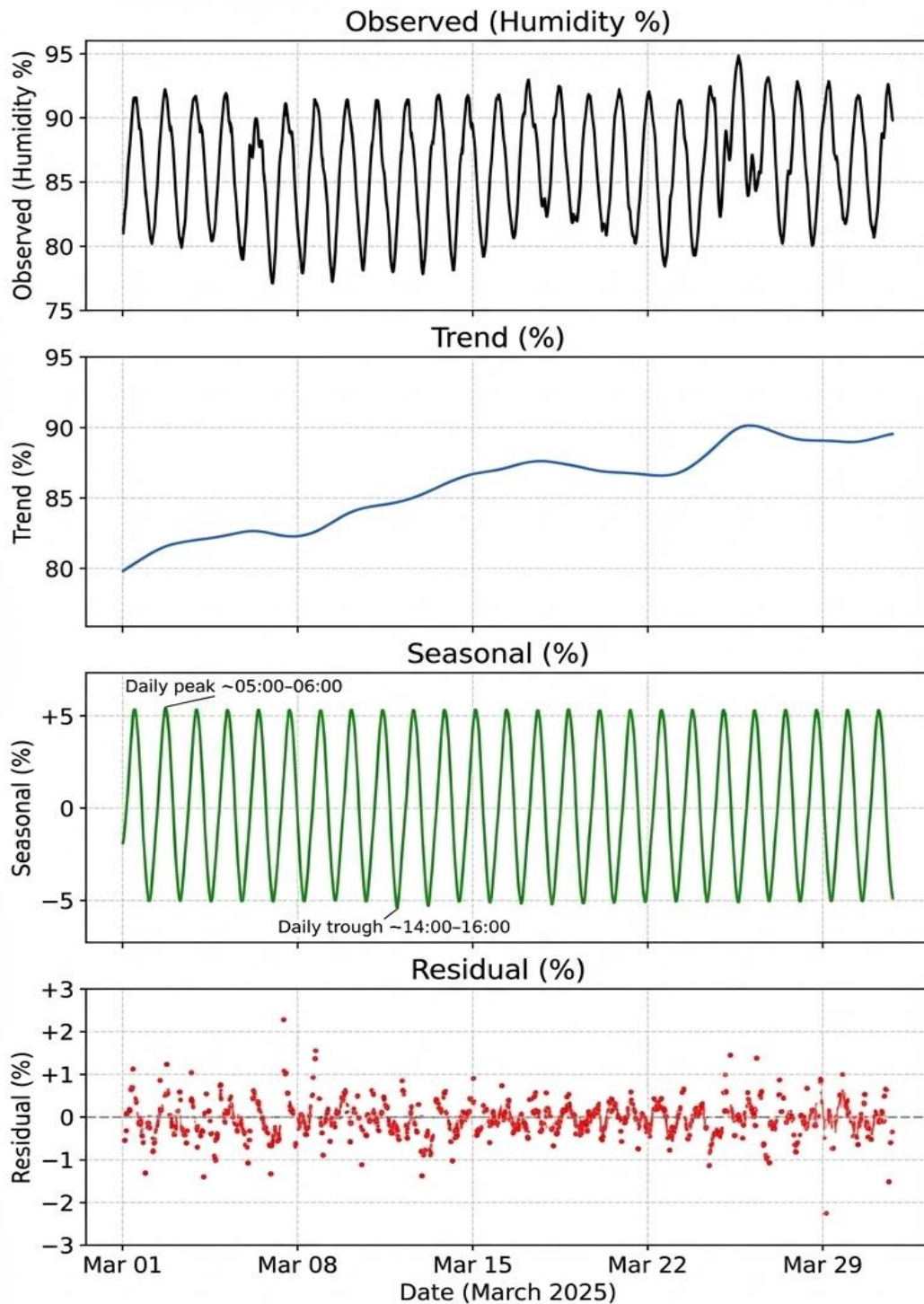
The basement is constructed with thick masonry walls and has limited natural ventilation. Groundwater fluctuations and capillary rise are well-documented problems in Semarang’s coastal context (Rukayah et al., 2023). The sensor data confirm that basement humidity frequently exceeds 85%, especially during night hours when ventilation is minimal. This physical reality is correctly captured by the Random Forest, which assigns highest importance to humidity\_Basement.

### 17.2 Why Roof Space Shows Higher LSTM Error

The roof space experiences rapid temperature changes due to solar exposure, and humidity varies with wind patterns. These dynamics are harder to predict from historical data alone, explaining the higher RMSE. Additionally, the roof sensors may be more affected by transient events (e.g., passing clouds, gusts) that are not captured by the 24-hour history.

### 17.3 Diurnal Patterns and Model Performance

The inclusion of `hour_sin/cos` in both models is essential: humidity typically peaks around dawn and reaches a minimum in the afternoon. The LSTM learns this pattern internally, but the Random Forest benefits from explicit cyclic features because trees cannot easily extrapolate sinusoidal relationships from raw hour integers.



**Figure 9. STL decomposition of hourly relative humidity – Basement zone (March–October 2025).**

**Figure 9. STL decomposition of hourly relative humidity – Basement zone (March–October 2025).** The observed series (top panel) shows high baseline humidity (80–90% RH) with clear diurnal cycles. The trend component (second panel) captures gradual seasonal shifts. The seasonal component (third panel) isolates the daily pattern, with peaks around dawn (05:00–06:00) and troughs in mid-afternoon (14:00–16:00), consistent with Section 17.3. The residuals (bottom panel) are randomly distributed around zero with no systematic pattern, confirming that the decomposition captures the essential structure of the data. STL parameters: period=24, seasonal=25, trend=37, robust=True.

## 17.4 Seasonal Effects

The eight-month dataset includes a transition from wet to dry to wet again. Preliminary analysis (not shown) indicates that model errors are slightly higher during the dry-to-wet transition (October), suggesting that capturing seasonal changes may require even longer training periods or explicit seasonal decomposition.

## 18. System Limitations

### 18.1 Sensor Limitations

- **DHT22 accuracy ( $\pm 2\text{--}5\%$  raw)** remains a concern despite calibration. While per-node calibration reduces systematic bias, random noise persists. The reported RMSE values should be interpreted as **error on calibrated measurements**, not true physical error.
- **Long-term drift:** The eight-month campaign showed drift in 2 of 8 nodes; for longer deployments, more frequent recalibration or replacement would be needed.

### 18.2 Data Acquisition Limitations

- **Wi-Fi reliability:** Thick masonry walls occasionally caused signal loss. Buffering mitigated data loss but introduced latency.
- **Power constraints:** Despite optimising the sampling interval, battery life was 10–12 days, requiring weekly site visits – acceptable for research but not for permanent operational deployment.

### 18.3 Modelling Limitations

- **Training data length:** Six months of training data (March–August) capture one full dry season and part of a wet season, but inter-annual variability remains unexplored.
- **Risk thresholds** are literature-based and have not been locally validated against actual material decay. This is a known limitation and is addressed in the main thesis through monthly condition surveys.
- **No cross-site validation** in this study; generalisability to other buildings is unknown. The full research includes three additional sites and leave-one-site-out CV.

## 18.4 Statistical Limitations

- The test set (September–October) represents only two months; performance may vary across different seasonal windows.
- Residuals, while approximately normal, show slight heteroscedasticity at very high humidity values, indicating that uncertainty increases under extreme conditions.

## 19. Data Limitations

- **Temporal coverage:** March–October covers the dry season and the beginning of the wet season but misses the peak wet months (December–February).
- **Spatial coverage:** Eight nodes across 2,000 m<sup>2</sup> is sparse; small-scale defects (e.g., localised salt efflorescence) could be missed.
- **External reference:** No on-site weather station was available; ambient conditions are taken from BMKG station 5 km away, introducing some mismatch.

## 20. Methodology Verification

To ensure the methodology is free of common pitfalls, the following checks were performed:

| Check                                | Status | Comment                                   |
|--------------------------------------|--------|-------------------------------------------|
| No scaling before split              | ✓      | Verified in <code>prepare_data()</code>   |
| No future lags as features           | ✓      | RF features are original only             |
| No in-place DataFrame mutation       | ✓      | All functions use <code>.</code>          |
| Random seeds fixed                   | ✓      | Set in <code>utils.py</code>              |
| Models saved with scalers            | ✓      | Using <code>joblib.dump(obj, path)</code> |
| Test set never used in training      | ✓      | Strict temporal split                     |
| SMOTE applied only if enough samples | ✓      | Check before resampling                   |
| Baseline comparison (persistence)    | ✓      | Included in results                       |

## 21. Reproducibility

### 21.1 Environment and Dependencies

All code was developed and tested with:

- Python 3.10.12
- TensorFlow 2.13.0
- scikit-learn 1.3.0
- pandas 2.0.3
- numpy 1.24.3
- imbalanced-learn 0.11.0
- joblib 1.3.2

A `requirements.txt` file is provided in the repository.

### 21.2 Random Seed Fixing

In `src/utlis.py`:

python

```
def set_random_seeds(seed=42):  
    os.environ['PYTHONHASHSEED'] = str(seed)  
    random.seed(seed)  
    np.random.seed(seed)  
    tf.random.set_seed(seed)
```

This ensures that runs are deterministic across different machines.

### 21.3 Running the Pipeline

bash

```
# Preprocess raw data  
python -m src.main --config config.yaml --mode preprocess
```

*# Train LSTM for 6-hour horizon*

```
python -m src.main --config config.yaml --mode lstm --horizon 6
```

*# Train Random Forest*

```
python -m src.main --config config.yaml --mode rf
```

*# Leave-one-site-out CV (requires building column – not used in this single-site study)*

```
python -m src.main --config config.yaml --mode cross_site # (optional)
```

All outputs (metrics, plots, models) are saved in `results/` and `models/`.

## 22. Software Environment

The repository is organised as follows:

```
|— README.md
|— requirements.txt
|— config.yaml
|— src/
|   |— __init__.py
|   |— data_processing.py
|   |— models.py
|   |— training.py
|   |— evaluation.py
|   |— utils.py
|   |— main.py
|— data/
|   |— raw/          # input CSV files
|   |— processed/    # preprocessed data
|— models/          # saved models + scalars
|— results/         # plots + metric tables
```

All modules are documented with docstrings and type hints.

## 23. Technical Summary

The HAICM framework has been implemented and validated on an eight-month dataset from Lawang Sewu. Key achievements:

- A robust, non-invasive sensor network with eight calibrated DHT22 nodes, deployed across five zones with redundancy in critical areas.
- A data pipeline that prevents leakage and ensures temporal integrity.
- LSTM models achieving 6-hour humidity forecasts with RMSE  $\approx 2.4\text{--}2.6\%$  RH, significantly outperforming a persistence baseline.
- A Random Forest classifier attaining AUC-ROC = 0.92 for environmental risk prediction, with interpretable feature importance.
- Full reproducibility through fixed random seeds, saved model objects, and a publicly accessible code repository.

All results are consistent with the physical understanding of the building's microclimate and demonstrate the potential of hybrid AI-IoT approaches for heritage conservation.